

OE-EC604A — Electronic Measurement & Measuring Instruments

Semester VI • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: Signal Analyzers & Generators

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Contents

1	Signal Analyzers	2
1.1	Wave Analyzers	2
1.2	Total Harmonic Distortion (THD)	2
1.3	Spectrum Analyzers	3
2	Signal Generators	3
2.1	Function Generators	3
2.2	Arbitrary Waveform Generators (AWG)	3

1. Signal Analyzers

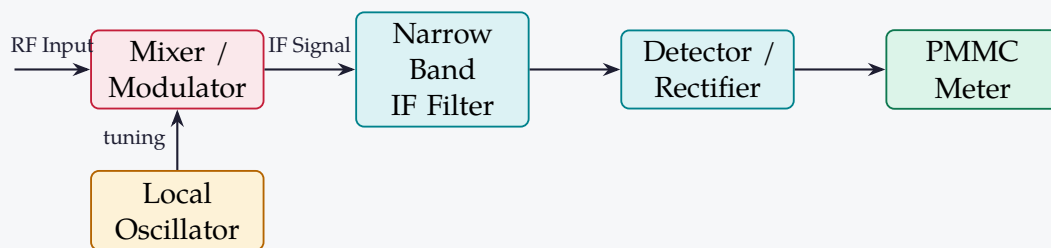
Signal analyzers resolve electrical signals into their individual frequency components, measuring amplitude, distortion, and spectral properties.

1.1 Wave Analyzers

A **Wave Analyzer** is an instrument designed to measure the amplitude of a single frequency component of a complex signal.

- **AF Wave Analyzer:** Uses a frequency-selective amplifier with a tunable active filter (Wein bridge or RC network) to isolate the target frequency.
- **Heterodyne Wave Analyzer (HF):** Operates on the superheterodyne principle. The input signal is up-converted to a high Intermediate Frequency (IF) using a tunable Local Oscillator and Mixer. A narrow-band bandpass filter extracts the target frequency, which is then rectified and indicated on a PMMC meter.

Heterodyne Wave Analyzer Block Diagram



1.2 Total Harmonic Distortion (THD)

Distortion analyzers measure the deviation of a waveform from a perfect sinusoid. When a pure sine wave passes through a non-linear device, harmonics are generated.

Harmonic Distortion Calculations

Let V_1 be the fundamental component amplitude, and $V_2, V_3, \dots, V_n$ be the amplitudes of the second, third, and higher order harmonics.

- **Individual Harmonic Distortion (D_n):**

$$D_n = \frac{V_n}{V_1} \times 100\%$$

- **Total Harmonic Distortion (THD):**

$$\text{THD} = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots + V_n^2}}{V_1}$$

- **THD relative to Total RMS Value (V_{rms}):**

$$\text{THD}_t = \frac{\sqrt{V_2^2 + V_3^2 + \dots + V_n^2}}{\sqrt{V_1^2 + V_2^2 + \dots + V_n^2}} = \frac{\sqrt{V_{\text{rms}}^2 - V_1^2}}{V_{\text{rms}}}$$

1.3 Spectrum Analyzers

Unlike wave analyzers that measure one frequency component at a time, a **Spectrum Analyzer** swept-tunes across a frequency band and displays the entire frequency spectrum (amplitude vs frequency) on a screen. - **Working:** A saw-tooth sweep generator sweeps the tuning voltage of a Voltage Controlled Oscillator (VCO). This VCO serves as the local oscillator for a superheterodyne mixer, sweeping the input signal past a narrow bandpass filter. The output is rectified and drives the vertical deflection of a CRT, while the saw-tooth generator drives the horizontal sweep, yielding the spectral display.

2. Signal Generators

Signal generators produce controlled electrical waveforms for testing, calibration, and troubleshooting.

2.1 Function Generators

A **Function Generator** produces multiple standard waveforms (sine, square, triangular) over a wide frequency range. - **Working Principle:** It utilizes an **integrator** and a **comparator (Schmitt trigger)** feed-back loop. The integrator charges/discharges a capacitor using constant current sources, yielding a linear triangular wave. The comparator toggles its output state at peak voltages, yielding a square wave. The triangular wave is fed through a diode-resistance wave-shaping network to approximate a low-distortion sine wave.

2.2 Arbitrary Waveform Generators (AWG)

An **AWG** generates custom, user-defined waveforms. It functions digitally: 1. The user defines the waveform shape mathematically or via sampling, storing the digital coordinates in a **Waveform Memory**. 2. A **Sampling Clock** drives an address counter to read out the digital words sequentially. 3. A high-speed **Digital-to-Analog Converter (DAC)** converts the digital words back into a continuous analog voltage. 4. A **Reconstruction Filter** smooths out the quantization steps.

Quick Revision — Module 2

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Key Distortion Formula

$$\text{THD} = \frac{\sqrt{V_2^2 + V_3^2 + \dots + V_n^2}}{V_1} \quad D_n = \frac{V_n}{V_1}$$

$$\text{Total AC Power RMS: } V_{\text{rms}} = \sqrt{V_1^2 + V_2^2 + V_3^2 + \dots}$$

One-Line Definitions

- **Wave Analyzer:** Frequency-selective voltmeter tuned to measure a single harmonic component.
- **Spectrum Analyzer:** Swept-frequency instrument displaying amplitude vs frequency across a band.
- **THD:** Total Harmonic Distortion; ratio of root-sum-square of all harmonics to fundamental amplitude.
- **Function Generator:** Signal source using integrator-comparator loops to yield sine, square, and triangle waves.
- **Sweep Generator:** Generator sweeping output frequency automatically across a range for response plotting.
- **AWG:** Arbitrary Waveform Generator; digitally synthesises arbitrary user-defined waveforms from RAM.

Frequently Asked / Viva Questions

Q1. Compare Wave Analyzers and Spectrum Analyzers.

A wave analyzer is a narrow-band selective voltmeter tuned manually to measure a single isolated frequency component. A spectrum analyzer automatically sweeps across a band of frequencies and displays the entire spectrum (amplitude vs frequency) in real-time.

Q2. An instrument measures a fundamental component $V_1 = 10 \text{ V}$ and harmonics $V_2 = 1 \text{ V}$, $V_3 = 0.5 \text{ V}$. Calculate the THD.

$$\text{Harmonic RMS: } V_h = \sqrt{V_2^2 + V_3^2} = \sqrt{1^2 + 0.5^2} = \sqrt{1 + 0.25} = \sqrt{1.25} \approx 1.118 \text{ V.}$$

$$\text{THD} = V_h/V_1 = 1.118/10 = 0.1118 \Rightarrow 11.18\%.$$

Q3. Explain the working principle of a basic Function Generator.

A function generator uses an integrator and a Schmitt trigger comparator. The integrator generates a triangular wave by charging a capacitor with constant current. The comparator detects the peaks and switches the charging current

Frequently Asked / Viva Questions (contd.)

direction, generating a square wave. A diode shaping network converts the triangle wave to a sine wave.

Q4. What is the role of the reconstruction filter in an Arbitrary Waveform Generator (AWG)?

The digital-to-analog converter (DAC) reads discrete memory coordinates and outputs a stair-step approximation of the waveform. The reconstruction filter acts as a low-pass filter, removing the high-frequency quantization steps to output a smooth, continuous analog signal.

Q5. What is heterodyning in HF wave analyzers?

Heterodyning is the mixing of an input high-frequency signal (f_s) with a variable local oscillator frequency (f_{lo}) in a non-linear mixer. This translates the target frequency component to a fixed intermediate frequency ($IF = |f_{lo} - f_s|$) where it can be filtered by a high-Q bandpass filter.

Q6. Explain the function of a Sweep Frequency Generator.

A sweep generator automatically sweeps its output frequency continuously across a defined frequency band. It is used to test the frequency response of amplifiers, filters, and RF components by synchronizing the sweep with an oscilloscope display.

Q7. What is a Capacitance-Voltage (C-V) meter and what is it used for?

A C-V meter measures the capacitance of a semiconductor junction (e.g., diode, MOS capacitor) as a function of an applied DC bias voltage. It is used to analyze doping profiles, oxide thickness, and interface trap densities.

Q8. What is harmonic distortion and what causes it?

Harmonic distortion is the presence of unwanted integer multiples (harmonics) of the fundamental frequency in a signal. It is caused by non-linearities in electronic components (like transistors, diodes, or saturated magnetic cores) within the signal path.

Q9. How does a Spectrum Analyzer generate its horizontal scale?

A saw-tooth ramp generator drives both the horizontal sweep plates of the display tube and the tuning voltage input of a voltage-controlled oscillator (VCO). This ensures that the horizontal position directly corresponds to the frequency of the local oscillator.

Q10. Define the term Form Factor in AC measurements.

Form factor is the ratio of the RMS value of a waveform to its average value (Form Factor = V_{rms}/V_{avg}). For a pure sine wave, it is 1.11.

Wave Analyzer vs Spectrum Analyzer Comparison

Parameter	Wave Analyzer	Spectrum Analyzer
Display Mode	Single meter deflection indicator.	Real-time CRT/LCD spectral plot.
Domain	Single isolated frequency point.	Entire selected frequency band.
Sweep	Manual tuning only.	Automatic swept-tuned VCO.
Complexity	Lower, behaves as a narrow tuner.	High, requires VCO and sweep generator.